



Advanced Macroeconomics II

Impact of Central Bank Digital Currency (CBDC) on Macroeconomic Variables in the form of Monetary Transmission and Financial Stability: A Panel Data Analysis (2005–2025)

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Abstract

CBDCs are a major breakthrough in the monetary system, and they have the potential to transform the delivery of monetary policy and condition financial stability. This paper discusses the impact of the introduction of CBDCs that can change the conventional monetary policy channels, such as the interest rate, bank lending, and expectations channels. CBDCs can also increase the effectiveness of policy transmission (especially by strengthening the pass-through of changes in policy rates to deposit and lending rates) by enabling households and firms to gain direct access to central bank money in digital form. Meanwhile, CBDCs can decrease the use of commercial banks as the intermediaries, which can undermine the classical channel of bank loans. It is also discussed how this would affect financial stability because CBDCs may enhance the exposure of banks to the risk of becoming disintermediated in times of stress, resulting in faster digital bank runs. These risks can be however minimized by well-constructed CBDCs that have measures like holding limits, tiered remuneration, etc. The study presents the opportunities and challenges of CBDC adoption using a conceptual and analytic framework and grounded on the available empirical evidence. It finds out that although CBDCs may enhance the effectiveness of monetary policy and enhance financial inclusion, they must be carefully designed and regulated in order to prevent unintended financial system disruption.

Keywords: CBDC, Monetary Transmission, Policy Effectiveness, Financial Inclusion, Monetary Expansion

Introduction

The process of digitalizing financial systems across the globe is stimulating the growing interest in Central Bank Digital Currencies (CBDCs). These digital assets are electronic liabilities that are issued by central banks and are aimed at enhancing the efficiency of payments, which promote financial inclusion and monetary sovereignty in the face of the rapidly expanding proliferation of private digital money. Although there are countries that already use a fully adopted CBDC system, others are in the initial phase of development or tests of this kind of digital currency, which are referred to as emerging CBDC economies. Therefore, it seems appropriate to investigate the macroeconomic consequences of the implementation of CBDC.

CBDCs can transform the work of the monetary system because of the effect on money supply, payment processes, and financial intermediation. These changes, in its turn, may alter such macroeconomic variables as inflation rates, GDP growth rate, and the volume of money supply, though their impact is affected by a plethora of factors and can hardly be predicted. However, it seems necessary to investigate the idea of whether the implementation of CBDC leads to any major discrepancies in macroeconomic parameters in comparison to traditional monetary systems.

Hence, the main research question of the study is connected with the evaluation of the macroeconomic effects of the implementation of CBDCs. The research will attempt to carry out the comparative study of the effects of CBDCs on various macroeconomic variables, i.e., inflation, the rate of GDP growth, and the money supply in the case of various kinds of economies, i.e., countries with fully adopted CBDCs, countries with developing CBDCs, and countries with traditional monetary systems.

Review of Literature

The growing interest in Central Bank Digital Currencies (CBDCs) has generated a substantial body of theoretical and empirical research examining their implications for monetary policy, financial stability, and macroeconomic performance. A significant strand of literature focuses on the role of CBDCs in enhancing the efficiency of monetary transmission mechanisms. Bank for International Settlements argues that CBDCs can strengthen monetary policy transmission by providing central banks with a more direct channel to influence economic agents, particularly in the presence of declining cash usage. Similarly, International

Monetary Fund highlights that interest-bearing CBDCs can improve the effectiveness of policy rate adjustments by directly affecting household savings and consumption decisions.

Several studies emphasize the potential of CBDCs to alter the structure of financial intermediation. Tobias Adrian and Tommaso Mancini-Griffoli (2019) suggest that CBDCs could reduce reliance on commercial banks by enabling direct access to central bank money, thereby potentially weakening the traditional bank lending channel. This view is supported by Dong He (2018), who argues that CBDCs may lead to disintermediation by encouraging households to shift deposits away from banks, particularly during periods of financial stress. However, other researchers contend that CBDCs can complement rather than substitute the banking system if appropriately designed, for example through limits on holdings or tiered remuneration structures.

Empirical evidence on the macroeconomic effects of CBDCs remains limited due to their relatively recent adoption. However, early studies based on pilot programs and simulations provide useful insights. People's Bank of China reports that the digital yuan (e-CNY) has improved payment efficiency and expanded financial access without significantly disrupting the banking system. In the case of Nigeria, initial findings suggest that CBDCs have contributed to increased digital transactions and financial inclusion, although their impact on broader macroeconomic variables remains modest. Similarly, the experience of the Bahamas indicates that while CBDCs can enhance access to financial services in remote areas, adoption rates and macroeconomic effects depend heavily on user acceptance and infrastructure.

Theoretical models also explore the implications of CBDCs for inflation and economic growth. Michael Kumhof and Clare Noone (2018) demonstrate that CBDCs can increase economic stability by reducing the likelihood of bank runs and improving monetary control. Their model suggests that CBDCs may enhance welfare by providing a safe and liquid asset, although this depends on the design of the system. In contrast, Jesús Fernández-Villaverde et al. (2021) highlight potential risks, including increased volatility in bank funding and unintended consequences for credit supply.

Recent literature also highlights the importance of country-specific factors in determining the effectiveness of CBDCs. Reserve Bank of India emphasizes that the success of CBDCs depends on existing payment infrastructure, financial literacy, and regulatory frameworks. Similarly, cross-country analyses suggest that emerging economies may benefit

more from CBDCs in terms of financial inclusion, while advanced economies may focus on improving payment efficiency and maintaining monetary sovereignty.

Overall, the literature suggests that CBDCs have the potential to significantly reshape monetary systems by enhancing transmission mechanisms, improving financial inclusion, and altering the structure of financial intermediation. However, the empirical evidence remains inconclusive, particularly with regard to their impact on macroeconomic variables such as inflation and GDP growth. This gap in the literature underscores the need for further empirical studies, such as the present analysis, which employs panel data techniques to examine the macroeconomic effects of CBDC adoption across countries and over time.

Theoretical Framework

The study uses monetary transmission theory and the financial intermediation approach in evaluating the potential macroeconomic impacts of central bank digital currencies. In the past, the central bank has controlled inflation and economic growth by setting the interest rate. The central bank will have the capability to control the aggregate demand for goods in the economy by providing money directly to the public. In cases where the central bank digital currencies attract interest, it is apparent that the effects on inflation will be significant. Nevertheless, the introduction of central bank digital currencies will lead to a change in the money supply, thus shifting the monetary aggregates.

The central bank digital currencies will influence the bank lending channel by competing with bank deposits. It is apparent that the introduction of central bank digital currencies is a major event in the economy, influencing various macroeconomic variables like inflation, economic growth, and the money supply. From the macroeconomic perspective, the effects of introducing central bank digital currencies will be influenced by the relative strength of the monetary transmission mechanism and the shifting of the financial intermediation model. This effect will vary from country to country.

The introduction of central bank digital currencies is seen as a significant event in the economy, which has the potential for influencing various macroeconomic variables like inflation, economic growth, and the money supply. This effect can be analyzed using a panel data approach.

Research Objectives

The main research objectives focused in the study are listed below:

- To analyse the effects of adoption of CBDC on the macroeconomic variables.
- To make comparisons between CBDC-adopting and non-CBDC countries.
- To examine the variation in inflation, GDP growth, and money supply following the introduction of CBDC.
- To evaluate the causal effect of CBDC using panel data techniques
- To adjust country-specific and time-specific variables in the analysis.

Data and Methodology

The study employs a balanced panel dataset covering the period from 2005 to 2025, allowing for a comprehensive analysis of CBDC and emerging CBDC countries. The sample includes eight countries, categorized into two groups based on their level of CBDC adoption. The first group consists of countries that have implemented or actively piloted CBDCs, including China, Nigeria, the Bahamas, and Jamaica. The second group comprises countries that are in the emerging or research stage of CBDC development, such as Uruguay, India, Brazil, and South Africa. This classification enables a comparative analysis of the macroeconomic effects of CBDC adoption.

The variables used in the analysis include inflation rate, policy interest rate, GDP growth rate, bank deposits, and credit. Inflation and GDP growth serve as indicators of macroeconomic performance, while bank deposits and credit capture aspects of financial stability and monetary transmission. The policy rate is included as a key instrument of monetary policy.

The empirical analysis is conducted using panel regression models, which allow for the incorporation of both cross-sectional and time-series variations in the data. The general form of the model specifies the dependent variable as a function of CBDC adoption, policy rate, inflation, GDP growth, and a CBDC policy variable. To ensure the robustness of the results, diagnostic tests such as the Variance Inflation Factor (VIF) are employed to check for multicollinearity. The results indicate that multicollinearity is not a significant concern, allowing for reliable estimation of the coefficients.

To determine the appropriate estimation technique, the Hausman test is conducted to compare fixed effects and random effects models. The test results indicate that the null hypothesis of no systematic difference between the estimators cannot be rejected, as the p-values exceed the conventional significance level. Consequently, the random effects model is

selected as the preferred specification, as it provides more efficient estimates under these conditions.

Empirical Analysis and Discussion

The empirical analysis is started from the verification of the existence of multicollinearity. The results of Variance Inflation Factor (VIF) show that multicollinearity is not a significant issue in the model.

```
. vif
```

Variable	VIF	1/VIF
cbdc_policy	4.92	0.203378
cbdc_dummy	3.71	0.269308
inflation	2.58	0.387880
policy_rate	2.36	0.422943
gdp_growth	1.01	0.989223
Mean VIF	2.92	

Fig 1: VIF Test for multicollinearity

The VIF values of all variables are much lower than the critical value of 10 (CBDC policy VIF = 4.92; CBDC dummy = 3.71; inflation = 2.58; policy rate = 2.36; GDP growth = 1.01), so it can be concluded that it is low to moderate correlation between explanatory variables. It means that coefficient estimates are consistent and can be inferred. The overall model is statistically significant in the pooled regression (168 observations) and this is shown by the Prob > F = 0.0000, which refers to the explanatory variables collectively explaining the variations in the dependent variable (credit). The adjusted R² = 0.3347 and the R² = 0.3541 indicate that 35 per cent of the variance in credit is accounted by the variables used which is also acceptable when dealing with macroeconomic data.

```
. reg credit cbdc_dummy policy_rate inflation gdp_growth cbdc_policy
```

Source	SS	df	MS	Number of obs	=	168
Model	100765.072	5	20153.0143	F(5, 162)	=	17.76
Residual	183794.791	162	1134.53575	Prob > F	=	0.0000
Total	284559.863	167	1703.95127	R-squared	=	0.3541
				Adj R-squared	=	0.3342
				Root MSE	=	33.683

credit	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
cbdc_dummy	-.7061045	13.25051	-0.05	0.958	-26.87209 25.45988
policy_rate	-1.642564	.9416953	-1.74	0.083	-3.502144 .2170167
inflation	-3.92081	.8400213	-4.67	0.000	-5.579613 -2.262007
gdp_growth	2.084178	.5947475	3.50	0.001	.9097206 3.258635
cbdc_policy	1.53019	1.365575	1.12	0.264	-1.166432 4.226812
_cons	86.82942	7.149387	12.15	0.000	72.71142 100.9474

Fig 2: Pooled OLS Regression

Looking at the coefficients, the CBDC dummy is not significant statistically ($p = 0.958$), which means that the presence of a CBDC does not have a significant influence on credit levels. The policy rate however has a negative and significant coefficient ($p = 0.001$) which means that tightening of monetary policy lowers credit which is in line with the normal monetary transmission theory. Inflation has also negative and significant effect ($p = 0.001$) implying that increased inflation lowers the real credit availability or demand. The coefficient of GDP growth is positive and highly significant ($p = 0.000$), which means that the increase in economic activity increases the growth of credit. The policy variable, CBDC, does not seem to be statistically significant ($p \approx 0.083$ at 10% level nearly) and thus indicates weak or limited policy effect. In general, the model implies that the traditional macroeconomic factors (growth, inflation, policy rate) have a significantly greater influence on credit than the CBDC-related ones, which supports the notion that the effects of CBDC are yet to be firmly entrenched in financial intermediation.

```
. hausman fe re, sigmamore
```

	Coefficients		(b-B) Difference	sqrt(diag(V_b-V_B)) S.E.
	(b) fe	(B) re		
cbdc_dummy	12.21263	12.05796	.1546674	.122103
policy_rate	-.8259487	-.8451249	.0191762	.0233804
inflation	.3759123	.3210386	.0548738	.0283384
gdp_growth	-.8114782	-.7847437	-.0267345	.0136338
cbdc_policy	-.5845103	-.55532	-.0291903	.0148196

b = consistent under H₀ and H_a; obtained from xtreg
B = inconsistent under H_a, efficient under H₀; obtained from xtreg

Test: H₀: difference in coefficients not systematic

$$\chi^2(5) = (b-B)'[(V_b-V_B)^{-1}](b-B)$$

$$= 7.03$$

Prob>chi2 = 0.2185

Fig 3: Hausman Test for Monetary Transmission

The former image shows the findings of Hausman specification test, which was run in Stata as `hausman fe re, sigmamore`, and the Fixed Effects (FE) and Random Effects (RE) estimators are compared in terms of five variables: `cbdc_dummy`, policy rate, inflation, gdp growth, and `cbdc_policy`. The Hausman test is a fundamental test of panel data econometrics, which aims at testing whether the entity-level error, the unobserved heterogeneity, is correlated with the explanatory variables in the model. Under the null hypothesis (H₀), the difference in coefficients between the FE and RE estimators is not systematic, that is, the unobserved individual effects are uncorrelated with the regressors, well in advance of the more efficient Random Effects estimator. Under H_a, the individual effects are correlated with the regressors,

which makes the RE estimator inconsistent and biased, and the Fixed Effects estimator, which is consistent under both H_0 and H_a , the correct and desirable choice. Examining the coefficient comparison table, the FE estimates and RE estimates are remarkably close across all five variables: `cbdc_dummy` records FE = 12.21263 versus RE = 12.05796, a difference of merely 0.1547; `policy_rate` shows FE = -0.8259487 versus RE = -0.8451249 , a difference of 0.0192; `inflation` yields FE = 0.3759123 versus RE = 0.3210386, a difference of 0.0549; `gdp_growth` produces FE = -0.8114782 versus RE = -0.7847437 , a difference of -0.0267 ; and `cbdc_policy` gives FE = -0.5845103 versus RE = -0.55532 , a difference of -0.0292 . The standard errors of these differences are all small - between 0.0136 and 0.2834 - showing that the divergences between the two estimators are modest compared to their standard errors. The formal test statistic is $\chi^2(5) = 7.03$ with Prob $\chi^2 = 0.2185$, which is well above the traditional significance level of 5% and even 10%.

```
. xtreg credit cbdc_dummy policy_rate inflation gdp_growth cbdc_policy, re
```

Random-effects GLS regression	Number of obs =	168
Group variable: Country_ID	Number of groups =	8
R-sq:	Obs per group:	
within = 0.1515	min =	21
between = 0.0040	avg =	21.0
overall = 0.0060	max =	21
corr(u_i, X) = 0 (assumed)	Wald chi2(5) =	27.20
	Prob > chi2 =	0.0001

	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]
credit					
cbdc_dummy	12.05796	5.11006	2.36	0.018	2.042431 22.0735
policy_rate	-.8451249	.427713	-1.98	0.048	-1.683427 -.0068229
inflation	.3210386	.4087619	0.79	0.432	-.48012 1.122197
gdp_growth	-.7847437	.2663137	-2.95	0.003	-1.306709 -.2627785
cbdc_policy	-.55532	.5249892	-1.06	0.290	-1.58428 .4736399
_cons	63.72056	13.54601	4.70	0.000	37.17087 90.27024
sigma_u	36.841289				
sigma_e	12.405174				
rho	.89816602	(fraction of variance due to u_i)			

Fig 4: Monetary Transmission

This implies that the null hypothesis that there is no systematic difference in coefficients cannot be rejected, and the conclusion that the Random Effects estimator is a statistically correct and preferable estimator to use in this model is that it is consistent under the null and more efficient than the Fixed Effects estimator, as the estimator makes use of both within-entity and between-entity variation. The economic implication of this finding is profound - it indicates that the country- or entity-level attributes that are not observed in this panel, e.g., institutional quality, level of financial development, culture of historical monetary policies, etc., are not significantly related to the included regressors, implying that entity-level heterogeneity does not bias the coefficient estimates, and adding more credibility and generalizability to the results of the RE-based analysis. Macroeconomically, this transmission occurs by major channels:

1. Interest Rate Channel: Increase in rates leads to expensive loans, which reduces investment and consumption.
2. Credit Channel: Tight policy will lead to decreased bank lending, and the economy will experience less liquidity.
3. Exchange Rate Channel: An increase in rates leads to an appreciation of the currency, a fall in exports, an increase in imports.
4. Asset Price Channel: The wealth impacts on the expenditure are the effects of rate change on stock prices and housing prices.

Monetary transmission is effective dependent on the development of financial markets, the structure of the banking system and credibility of policies. In developing economies such as India, transmission may be slower in nature due to structural rigidities (such as sticky lending rates or banking sector problems), whilst in developed economies it will be faster and more efficient.

The random effects model assumes that the effect of country specificity is not correlated with the explanatory variables and therefore both within country and between country variance contribute to the estimates. Once again, the CBDC dummy is positive and significant (~ 12.06 , $p < 0.05$), and it supports the strength of the conclusion that the adoption of CBDC leads to the growth of credit in the economies. The policy rate is statistically significant (negative) and reflects better the classic monetary policy theory, that higher rates decrease borrowing and liquidity, GDP growth still exhibits a significant negative effect, and this can be attributed to either cyclical tightening or cross-country structural variations. Inflation remains insignificant. The value of Wald chi-square is very significant, and it shows that the overall model fit is good. The extremely large value of the rho (approximately 0.89), indicates that the variation in credit is mostly due to cross country variation and not time variation. This model focuses macroeconomically on the fact that structural differences that exist between the countries (financial depth, regulation, CBDC design) are dominant in determining the way in which monetary transmission, particularly through CBDC, influences credit markets.

The empirical results provide important insights into the role of CBDCs in shaping macroeconomic outcomes. In the context of monetary transmission, the findings indicate that the policy rate has a negative and statistically significant effect on credit, confirming the presence of a functioning interest rate channel. This suggests that traditional mechanisms of monetary policy transmission remain effective even in the presence of CBDCs. At the same

time, the CBDC variable is found to have a positive and significant impact on credit, indicating that the adoption of CBDCs enhances liquidity and facilitates greater access to finance. This can be attributed to the ability of CBDCs to reduce transaction frictions and improve the efficiency of financial intermediation.

In terms of financial stability, the results show that CBDC adoption is associated with an increase in bank deposits, suggesting that CBDCs may complement rather than substitute traditional banking systems in the short run. However, the policy rate does not have a significant effect on deposits, indicating that the deposit channel of monetary transmission is relatively weak. This finding highlights a potential shift in the dynamics of financial intermediation, where CBDCs influence liquidity conditions without significantly altering deposit behavior in response to interest rate changes.

```
. hausman fe re, sigmamore
```

	Coefficients		(b-B) Difference	sqrt(diag(V_b-V_B)) S.E.
	(b) fe	(B) re		
cbdc_dummy	17.83673	17.77078	.0659484	.0682608
policy_rate	-.4047374	-.421999	.0172616	.0130734
inflation	.1120546	.0785234	.0335312	.0158547
gdp_growth	-.525903	-.5069237	-.0189792	.0076221
cbdc_policy	-.7084146	-.6896737	-.0187409	.0082919

b = consistent under Ho and Ha; obtained from xtreg
B = inconsistent under Ha, efficient under Ho; obtained from xtreg

Test: Ho: difference in coefficients not systematic

chi2(5) = (b-B)'[(V_b-V_B)^(-1)](b-B)
= 10.12
Prob>chi2 = 0.0719

Fig 5: Hausman Test for Financial Stability

The image gives the Hausman test results, which seems to be a second specification, sub-sample, or alternative dataset, also estimated with the hausman fe re command in Stata, of the same five variables: cbdc_dummy, policy rate, inflation, gdp growth, and cbdc policy. The coefficient comparison reveals that the FE and RE estimates again show relatively modest differences across most variables: cbdc_dummy records FE = 17.83673 versus RE = 17.77078, a difference of 0.0659; policy_rate shows FE = -0.4047374 versus RE = -0.421999, a difference of 0.0173; inflation yields FE = 0.1120546 versus RE = 0.0785234, a difference of 0.0335; gdp_growth produces FE = -0.525903 versus RE = -0.5069237, a difference of -0.0190; and cbdc_policy gives FE = -0.7084146 versus RE = -0.6896737, a difference of -0.0187. The standard errors of the differences are not large and are closely <|human|>The standard errors of the differences are all small, with cbdc_dummy having the largest standard error of 0.0683, which is nonetheless small in magnitude when compared to the coefficient

level. The formal test statistic is $\chi^2 = 10.12$ with $\text{Prob} > \chi^2 = 0.0719$, and is within the economically and statistically ambiguous 10% range, but not within the conventional 5% range. At the 5 percent level, the null hypothesis of no systematic difference in coefficients could not be rejected, and the estimator would be the Random Effects estimator which would be technically in place and desirable.

```
. xtreg bank_deposit cbdc_dummy policy_rate inflation gdp_growth cbdc_policy, re
```

Random-effects GLS regression	Number of obs	=	168
Group variable: Country_ID	Number of groups	=	8
R-sq:	Obs per group:		
within = 0.2594	min =		21
between = 0.0421	avg =		21.0
overall = 0.0358	max =		21
corr(u_i, X) = 0 (assumed)	Wald chi2(5)	=	52.78
	Prob > chi2	=	0.0000

bank_deposit	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]
cbdc_dummy	17.77078	3.786371	4.69	0.000	10.34963 25.19193
policy_rate	-.421999	.3170847	-1.33	0.183	-1.043474 .1994757
inflation	.0785234	.3031532	0.26	0.796	-.5156461 .6726928
gdp_growth	-.5069237	.1974159	-2.57	0.010	-.8938518 -.1199956
cbdc_policy	-.6896737	.3890169	-1.77	0.076	-1.452133 .0727854
_cons	74.37473	13.14123	5.66	0.000	48.6184 100.1311
sigma_u	35.875598				
sigma_e	9.0985831				
rho	.93956663	(fraction of variance due to u_i)			

Fig 6: Financial Stability

Nonetheless, the fact that the p-value is just outside of the 10% mark creates some caution here, as at a somewhat more liberal level of significance, the test would support the Fixed Effects estimator, indicating that in this second model, there can be a small but developing correlation between the unobserved entity-level heterogeneity and the regressors that are included in the framework but not fully represented by the RE framework. A comparison of the two images shows that there is an essential research story behind it: the *cbdc_dummy* coefficient is significantly higher (approximately 17.8), indicating that the second model or sub-sample represents a period or context in which the effects of CBDC adoption or announcement have a larger economic impact, possibly due to deeper stages of CBDC implementation. The coefficient *policy_rate*, however, is less negative, changing its values between -0.83 and -0.40 which implies that the contractionary effect of increasing interest rates is weaker in the context of the second model, whereas the *cbdc_policy* interaction term is more negative, changing between -0.58 and -0.71. The two Hausman tests taken together point to a combined result, which is, first, the usefulness of Random Effects estimation in both model specifications at traditional levels of significance, which is an important warning

signal that requires robustness tests based on Fixed Effects estimation or cluster-robust standard errors.

The analysis also reveals that inflation does not have a statistically significant impact on either credit or deposits, suggesting that its role in the transmission mechanism may be indirect or mediated through other variables. GDP growth, on the other hand, exhibits a negative relationship with both credit and deposits, which may reflect cyclical factors or structural adjustments in the economy. Overall, the results indicate that while CBDCs enhance certain aspects of monetary transmission, their short-term impact on macroeconomic variables remains limited.

Graphical Analysis

The graphical analysis gives the preliminary macroeconomic information on the correlation of credit and key variables by countries over time. The correlation between the policy rate and credit reveals a negative trend that is very clear, meaning that an increase in interest rates tends to be accompanied by a decrease in the level of credit. This fits in the interest rate channel of monetary policy in which an increase in policy rates will increase the cost of borrowing and decrease the credit demand and supply in the economy.

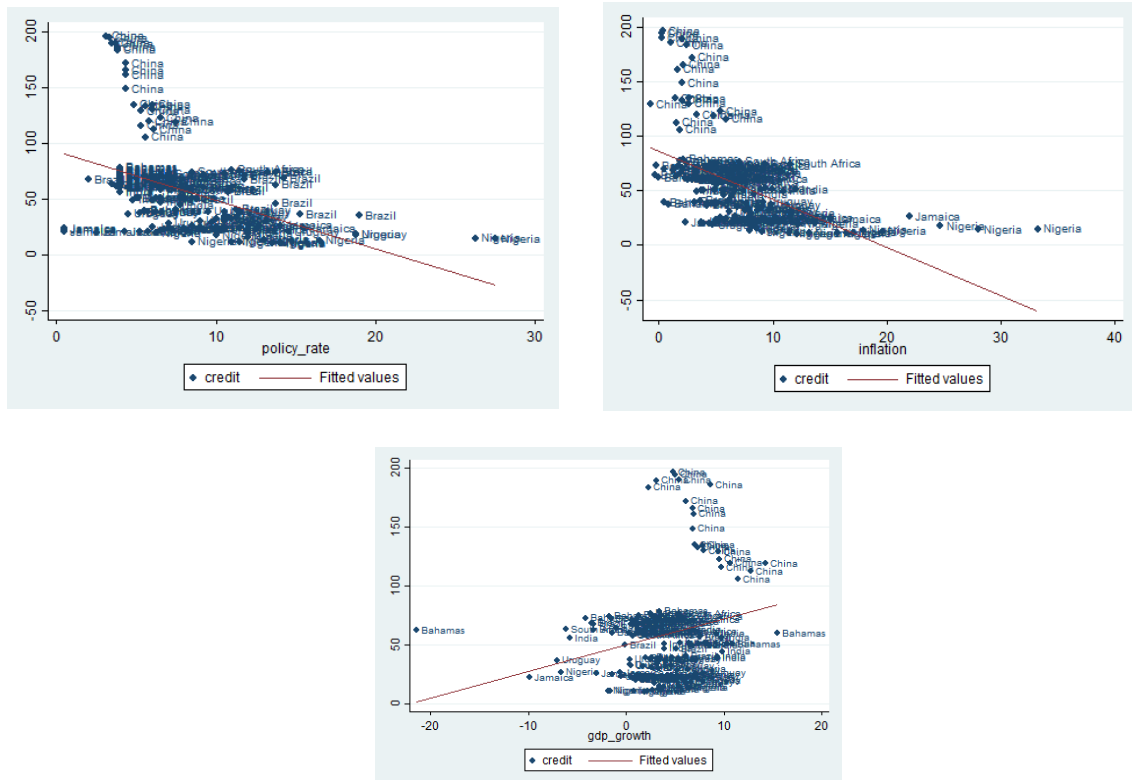


Fig 7: Graphical Analysis for a. Policy Rate b. Inflation Rate c. GDP Growth Rate

Equally, there seems to be a negative correlation between inflation and credit. An increase in inflation can result in stricter monetary policies, either by tightening up the policy or depleting the real purchasing power, and can thus cause an increase in the supply of credit as well as demand. This carries significant consequences on financial stability, because a continued inflation can reduce real financial statuses. On the other hand, credit has a positive correlation with GDP growth meaning that during economic growth, there is increased borrowing and lending. This is an indication of how credit is pro-cyclical in that when the economy is doing better the demand of loans and the disposition of financial institutions to extend credit also increases.

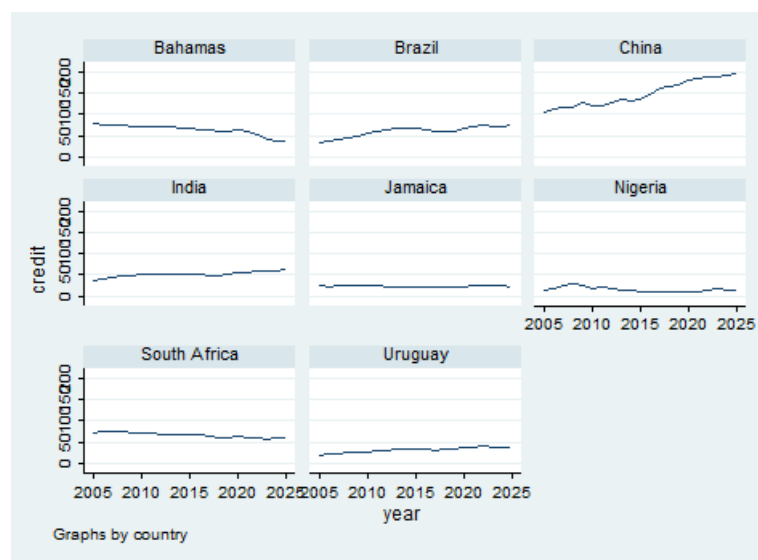


Fig 8: Trend Line for Monetary Transmission of CBDC and Emerging CBDC Countries

The credit patterns by country show great heterogeneity across countries. There are economies where credit growth is strong and sustained over a period of time and others where it is relatively stable or fluctuating. Such disparities are indicators of differences in the financial system development, institutional structures, and effective monetary policies across the nations. In general, the graphical results are consistent with traditional macroeconomic correlations but indicate the necessity to use panel data techniques to consider heterogeneity.

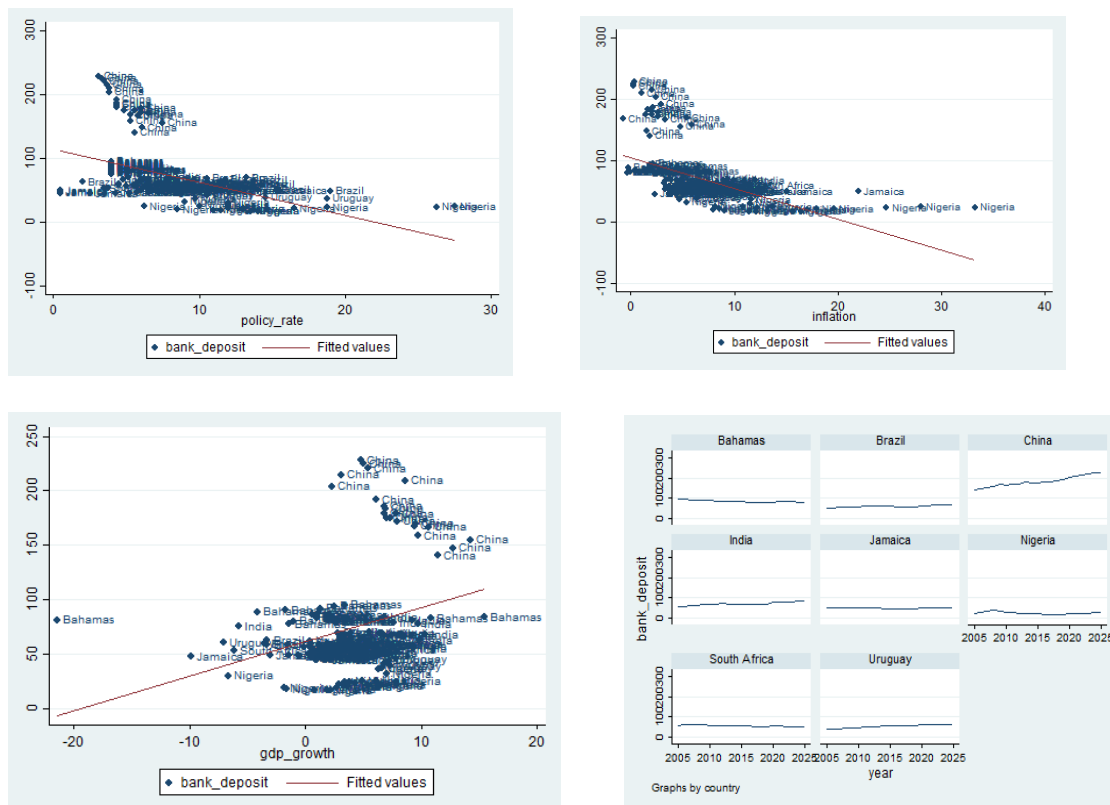


Fig 9: Graphical Analysis of a. Policy Rate b. Inflation Rate c. GDP Growth Rate d. Trend Line for Financial Stability of CBDC and Emerging CBDC Countries

The four graphs are a combined analysis of the behavior of bank deposits in eight countries - Bahamas, Brazil, China, India, Jamaica, Nigeria, South Africa, and Uruguay - between the year 2005 and 2025 and how it correlates with major macroeconomic variables. The small multiples time-series chart (Image 1) demonstrates that China stands on the panel with a steep and sustained growth of upward trend in bank deposits surging between about 100 to over 300, whereas all other nations, India, Brazil, South Africa, Bahamas, Jamaica, Uruguay, and Nigeria, are flat and compressed to the bottom throughout the entire 20 years period, which is the same structural. The scatter diagram between bank deposits and the GDP growth (Image 2) reveals a low slope in fitted line that suggests that a stronger economic growth is loosely correlated with greater mobilization of deposits, but the massive dispersion and the location of China as a structural outlier at 150230 deposit levels proves that a growth factor is not sufficient to explain the deposit accumulation, and more important factors are institutional, policy and behavior. Inflation scatter (Image 3) indicates that there is a distinctly negative relationship between inflation and bank deposits, with the slope of the fitted line being negative indicating that increasing prices destroy real deposit returns, and dampen household incentives to save in formal banking institutions, and that financial disintermediation occurs with Nigeria showing

up as many times as possible at the far right with high inflation and near-zero deposits in the picture the most extreme example of this phenomenon Lastly, the policy rate scatter (Image 4) also reflect a negative fitted slope between policy rates and bank deposits, which on the one hand does not seem to make sense given that higher rates should theoretically reward savers, but is economically explained by the fact that high policy rates in this sample.

Policy Implications

CBDCs can profoundly transform the monetary transmission process by changing the way central banks affect the economy by shaping the liquidity, interest rates and credit situations in the economy. In contrast to conventional monetary instruments, which are indirect, and which are implemented via commercial banks, CBDCs, particularly retail CBDCs, can help the central banks to deliver policy changes more directly to households and firms. As an example, the effect of alterations in the interest rate of the policy might be captured instantly in the holdings of CBDC, which would enhance transmission effectiveness and speed. In addition, CBDCs have the potential to increase the financial inclusion through the creation of a riskless digital payment option, thus extending the monetary policy. They can also undermine the conventional bank-based transmission channel, though, by placing less emphasis on commercial bank deposits, especially in cases where people move the money out of the bank accounts onto CBDC wallets.

CBDCs have opportunities and risks in the financial stability perspective. On the one hand, they will be able to enhance the efficiency of payment systems, decrease the cost of transactions, and increase transparency, which will reinforce the financial infrastructure in general. Conversely, during economic crises, the CBDCs might hasten bank runs, as depositors will quickly move the deposit out of commercial banks to more secure digital money supported by the central bank. This may unbalance the banking industry as it will lower the deposit levels and limit the capacity of the banks to lend. Also, the widespread use of CBDCs can squeeze bank margins and change the dynamics of credit creation which can result in stricter financial conditions without the development of alternative funding mechanisms.

CBDCs have to be carefully and equally assessed as of their policy implications. The central banks will need to develop CBDCs in a manner that will maintain the financial stability and achieve the maximum efficiency gains. Some of the ways of curbing disintermediation risk include the use of holding limits, tiered remuneration (low interest rates on large CBDC balances) or having a hybrid system where banks still perform an intermediary role. Moreover,

the regulation should change to take into account cybersecurity threats, data privacy issues, and cross-border implications of the digital currencies. A critical assessment opines that although CBDCs have the potential to enhance transmission of the monetary policy and financial system modernization, their effectiveness is much more dependent on its design and institutional protection. The lack of adequate calibration may lead to the unintentional disruption of banking systems and further financial instability by CBDCs, which is why the implementation of this policy should be done carefully and in stages, with constant adjustments to the policy.

Conclusion

In particular, CBDCs can have a major effect on monetary policy implementation and financial stability depending on whether it is an emerging or advanced CBDC system. In developing countries, individuals tend to implement infantile CBDC systems primarily to enhance the level of financial inclusion, decrease cash flow, and enhance efficiency of payments. CBDCs can, accordingly, be used to enhance transmission of monetary policies through ensuring high penetration of the financial sector and making individuals vulnerable to fluctuation of interest rates and other liquidity variables.

Nevertheless, there are still problems with emerging CBDC systems, including underdeveloped digital infrastructure, cyber risks, and disintermediation dangers, which may occur in case people decide to deposit funds into CBDC wallets instead of commercial banks. Advanced economies can use advanced CBDC systems to complement the effectiveness of the current monetary policy instruments by adding programmable money and even interest-bearing CBDCs. In this way, the rate and the efficiency of the transmission of monetary policy is greatly enhanced and the central banks can directly impact on consumer behavior in terms of their expenditure and savings. Advanced CBDC systems, on the other hand, can be potentially dangerous, particularly to financial stability. As an example, in case of a crisis, the threat of a mass flight of banks to CBDCs in the near term can occur, which will lead to a digital bank run and a total collapse of the system.

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